

AI Energy Demand and the Economic Case for Space Data Centers

A transparent, interactive model: methodology and assumptions

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[!] **AI-generated, under verification.** This model and report were generated by Claude Opus 4.8 (Anthropic) under the supervision of A. Golkar and are currently undergoing verification. Do not trust the results unless manually verified. Equations below describe what the code computes; they are not independently peer-reviewed.

1. Purpose and scope

The model estimates future electricity demand from artificial intelligence, compares it to the electricity the Earth can supply, identifies the point at which demand outruns supply, and evaluates whether the resulting gap could be served more cheaply from orbit than from additional ground buildout. It is deliberately transparent: every assumption is an adjustable parameter, and the simulation is a pure function of those parameters. The aim is a tool for structured reasoning under deep uncertainty, not a point forecast.

2. Model architecture

Demand is built bottom-up from three segments evaluated annually from a base year to a horizon year: (i) consumers, (ii) corporate knowledge workers, and (iii) autonomous agents. Each segment converts a population of users into electricity via a per-user query rate and a per-query energy. Supply is global electricity generation growing at a chosen rate; AI demand is stacked on top of non-AI demand (homes, industry, transport) and the total is compared against generation. A breakeven year and an Earth-versus-space cost comparison are derived from the resulting trajectories.

3. Demand model

3.1 Logistic adoption

Each human-bounded segment follows a logistic (S-curve) from a starting level a_0 toward a ceiling L at rate r , in years t after the base year:

$$A(t) = L / (1 + \exp(-r (t - t_0))), \quad t_0 = \ln((L - a_0)/a_0) / r$$

3.2 Population base

Population P and the knowledge-worker base grow geometrically at rate g :

$$P(t) = P_0 (1 + g)^t$$

3.3 Consumer and corporate segments

Consumer users are the population times consumer adoption; corporate users are the knowledge-worker base times corporate adoption. Both are bounded by human headcount, so both saturate.

$$\begin{aligned} N_{\text{consumer}}(t) &= P(t) \cdot A_{\text{consumer}}(t) \\ N_{\text{corp}}(t) &= W(t) \cdot A_{\text{corp}}(t) \end{aligned}$$

3.4 Autonomous agents (decoupled from headcount)

Agents are modelled as a compute-driven fleet whose saturation is expressed as agents-per-human against the base population. This is a transparent sizing of the ceiling; the fleet can far exceed the number of people. The fleet itself follows a logistic toward that ceiling:

$$C_agents = (agents\ per\ human) \cdot P0$$

$$N_primary(t) = logistic(fleet0, C_agents, r_agent, t)$$

Recursive sub-agents multiply the live fleet. If each agent spawns Z sub-agents over D generations, the total instance count is the geometric series $1 + Z + Z^2 + \dots + Z^D$:

$$M(Z, D) = (Z^{(D+1)} - 1) / (Z - 1) \quad (M = D+1 \text{ when } Z = 1)$$

$$N_agents(t) = N_primary(t) \cdot M(Z, D)$$

M grows exponentially in depth D , so this term can dominate all others. It is disabled by default ($D = 0$, hence $M = 1$) so the central case is not inflated.

3.5 Energy intensity: efficiency versus Jevons rebound

Per-query energy evolves under two opposing forces. Hardware and algorithms reduce energy per query at rate e (negative); a Jevons rebound R re-spends a fraction of those savings on more and heavier work. The net trend applied to the base energy w_0 is:

$$net = (e/100) \cdot (1 - R/100)$$

$$w(t) = w0 \cdot (1 + net)^t$$

At $R = 100\%$ the forces cancel (net-flat per-query energy); above 100% demand outpaces efficiency; below 100% efficiency lowers demand over time.

3.6 Segment energy

For a segment with N (billions of) units, q queries per unit per day, per-query energy w (Wh), agent intensity multiple m (1 for humans), facility PUE, and training overhead h , annual energy in TWh is:

$$E = N \cdot 10^9 \cdot q \cdot 365 \cdot w \cdot m \cdot PUE \cdot (1 + h/100) / 10^{12}$$

$$D(t) = E_consumer + E_corp + E_agent$$

4. Supply and the demand stack

Global generation S grows geometrically. Electricity is consumed as it is generated, so the grid is essentially fully used today; there is no idle surplus waiting for AI. The model therefore tracks non-AI demand B (homes, industry, transport) with its own growth rate, and treats AI demand D as stacked on top of it. Total electricity demand is the sum, compared against generation:

$$S(t) = S0 \cdot (1 + gs)^t \quad (\text{generation})$$

$$Pop(t) = Pop0 \cdot (1 + gp)^t \quad (\text{population})$$

$$B(t) = b \cdot Pop(t) \cdot (1 + gc)^t \quad (\text{non-AI demand, } b = \text{per-capita use})$$

$$D_total(t) = B(t) + D(t) \quad (\text{non-AI} + \text{AI})$$

Non-AI demand B is per-capita electricity use b times population, so it scales with the population LEVEL, with population growth gp , and with a per-capita electrification trend gc . The entered base-year total (~30,600 TWh) fixes b at ~3,700 kWh per person against the 8.3B reference. It is set close to current generation at the base year, so AI begins as a thin sliver on top of an almost-full grid. AI can grow only into the margin that generation opens over non-AI demand. If non-AI demand grows as fast as or faster than generation, that margin never opens and AI can grow only by displacing other uses.

Units (validation note). All quantities are ELECTRICITY, measured in TWh of generation and consumption. Global electricity generation was about 30,664 TWh in 2024 (Ember), the figure used here. This is deliberately NOT total primary energy, which was about 180,000 TWh in 2024 on the substitution method (Our World in Data, Energy Institute). Primary energy counts oil, gas, and coal burned for transport, heat, and industry, and

inflates non-fossil electricity to a fossil-equivalent; roughly only a fifth of final energy is electricity. AI data centers consume electricity, so electricity generation is the binding supply. Benchmarking AI against the larger primary-energy figure would understate the constraint, because most primary energy cannot be delivered to a data center as electricity without first building new generation, which is exactly what supply growth represents.

5. Breakeven definition

The breakeven is the first year total demand (non-AI plus AI) exceeds generation: the point at which the world cannot produce enough electricity to power both the rest of the economy and AI. It is the physically grounded signal that new capacity is needed.

$$t^* = \min \{ t : B(t) + D(t) > S(t) \}$$

6. Earth versus space economics

The unserved gap is total demand beyond generation. Capacity is sized against the largest gap over the horizon (efficiency trends can make demand peak then ease, so the final year is not always the binding year). The gap energy is converted to an average continuous power:

$$\begin{aligned} \text{gap}(t) &= \max(0, B(t) + D(t) - S(t)) \\ W_{\text{gap}} &= \text{gap_peak} \cdot 10^{12} / 8760 \quad [\text{W}] \end{aligned}$$

Closing it on Earth costs the gap power times an all-in build cost per watt. In orbit, generation mass scales down with the orbital solar advantage (a panel in the right orbit is several times more productive), while compute and bus mass (~half) do not; the mass is priced via the launch cost and a fixed non-launch hardware cost:

$$\begin{aligned} \text{Earth\$} &= W_{\text{gap}} \cdot \text{cost_earth} \\ \text{kg/kW_eff} &= \text{kgPerKw} / \text{solarMult} + 0.5 \cdot \text{kgPerKw} \\ \text{Space\$}/W &= (\text{kg/kW_eff} / 1000) \cdot \text{launch\$}/\text{kg} + \text{nonLaunch\$}/W \\ \text{Space\$} &= W_{\text{gap}} \cdot \text{Space\$}/W \end{aligned}$$

Setting the two equal and solving for launch price gives the threshold at which orbit becomes cheaper than the ground, the number the business case hinges on:

$$\text{launch*}_\$/\text{kg} = (\text{cost_earth} - \text{nonLaunch\$}/W) / (\text{kg/kW_eff} / 1000)$$

7. Parameters, defaults, and benchmarks

Each parameter and growth rate is anchored to a reputable benchmark where one exists; the source column gives the figure and reference. Items marked "est." have no published benchmark and are reasoned estimates. The agents-per-human value and recursion depth are the dominant swing variables and are deeply uncertain; treat them as one point in a wide range, not a forecast.

Parameter	Default	Basis / source
Population & time		
World population	8.3 B	Worldometer / UN, mid-2026
Population growth	0.8 %/yr	UN WPP; held constant
Horizon	2026-2050	Modelling choice
Agents per human (ceiling)	7.5 /person	Anchor ~50 per knowledge worker; swing variable, est.
Consumer		
Adoption today / ceiling	0.12 / 0.90	~1B of 8.3B use a chatbot (OpenAI, Feb 2026)
Adoption speed	0.35 /yr	Estimate
Queries / user / day	12	Estimate from OpenAI usage disclosures
Corporate		
Knowledge workers	1.25 B	Gartner: >1B knowledge workers worldwide
Adoption today / ceiling	0.10 / 0.85	McKinsey: 65% (2024) to 72% (2025) of orgs use genAI; ~10% deeply augmented
Adoption speed	0.40 /yr	McKinsey: 65% to 72% of orgs in ~1 yr
Queries / worker / day	60	est.
Autonomous agents		
Fleet today	0.05 B	est.; nascent
Fleet growth	0.45 /yr	Gartner: enterprise agent adoption 25% to 50% by 2027
Queries / agent / day	400	est.; 24/7 operation
Agent energy multiple	10x	Reasoning/agentic order of magnitude (Epoch AI)
Sub-agents per agent (Z)	3	speculative; only active when D > 0
Recursion depth (D)	0	off by default; exponential wild-card
Energy intensity		
Energy per query	0.30 Wh	0.24-0.34 Wh median (Epoch AI; OpenAI; IEEE)
Facility PUE	1.25	Uptime 2024: avg 1.56, hyperscale <=1.4
Training overhead	25 %	est. (10-40% range)
Hardware efficiency	-20 %/yr	Epoch: leading HW ~40%/yr more efficient; -20% conservative blend
Jevons rebound	100 %	neutral midpoint (net-flat per query); economics
Supply & non-AI demand		
Global generation	30,664 TWh	Ember Global Electricity Review 2025 (2024)
Supply growth	3.0 %/yr	IEA/Ember (4% in 2024; 2.6% 2010-23 avg)
Non-AI demand today	30,600 TWh	~ current generation; AI is ~0.1% today (Ember)
Non-AI per-capita growth	1.2 %/yr	IEA; total adds population growth (~2%/yr total)
Earth vs space		
Earth build cost	3.0 \$/W	power only; AI facilities \$20-40/W incl. compute (Epoch/Lazard)
Launch cost	1,000 \$/kg	Falcon 9 ~\$2,720/kg; Starship target <\$100/kg; needs ~200-500
Mass per kW (space)	6 kg/kW	est.; panels+radiators+bus+chips

Parameter	Default	Basis / source
Orbital solar gain	8x	Google Project Suncatcher
Space hardware cost	2.0 \$/W	est., excl. launch

8. Key findings (default scenario)

Under the default assumptions, AI electricity demand rises from about 0.1% of global generation in 2026 to roughly 0.6% in 2030, 28% in 2040 and 67% in 2050. Because non-AI demand already consumes almost the entire grid, total demand (non-AI plus AI) crosses generation in 2037, after which an unserved gap opens, reaching on the order of 28,000 TWh (a few thousand GW of continuous power) by 2050. The agent segment dominates AI demand; consumer and corporate use are small by comparison. Space becomes cheaper than additional Earth buildout once launch cost falls below roughly 270 \$/kg, consistent with the published Starcloud (~500 \$/kg) and Google Suncatcher (<200 \$/kg) targets.

Robust vs contingent. Robust across essentially the full plausible range of the agent ceiling: total demand outruns generation in the late 2030s, opening a multi-thousand-gigawatt gap. Contingent and genuinely uncertain: whether AI alone ever exceeds total global generation, which is governed almost entirely by agents-per-human and the recursion depth. The model affirms the first and leaves the second explicitly open.

9. Validation against known benchmarks

At the default settings the model reproduces independently published present-day figures. The first four rows are exact anchors or direct cross-checks; the AI-electricity row is AI-only and so sits below the IEA's all-data-center figure, of which AI is the fast-growing subset. The long-run trajectory beyond ~2030 is a steerable scenario, not a benchmarked forecast.

Benchmark	This model	Published	Source
Global electricity generation, 2024	30,664 TWh	30,664 TWh	Ember
World population, 2026	8.3 B	8.3 B	UN / Worldometer
Energy per AI query	0.30 Wh	0.24-0.34 Wh	Epoch AI; OpenAI
ChatGPT inference (2.5B queries/day)	~0.27 TWh/yr	~0.3 TWh/yr	OpenAI disclosures
AI electricity, 2026 (AI only)	~38 TWh	~40-85 TWh (AI ~10-20% of 415 TWh all DC)	IEA

10. Limitations and caveats

(1) Supply is a smooth exponential and ignores grid build-out friction, siting, and intermittency. (2) Demand saturation is imposed through logistic ceilings and a fixed agent ceiling; the true ceiling is economic and unknown. (3) Sub-agent recursion is a simple geometric tree; real sub-agents differ in lifetime and load. (4) Per-query energy, query rates, and the agent intensity multiple are estimates with wide uncertainty. (5) The space cost model is a first-order capex comparison and omits operations, cooling at scale, radiation hardening, latency, debris, and regulatory cost. (6) All figures are AI-generated and pending manual verification. The model's value is comparative and exploratory, not predictive.

11. Selected sources

- Ember, Global Electricity Review 2025 (2024 electricity generation ~30,664 TWh).
- Our World in Data / Energy Institute, primary energy ~180,000 TWh (2024, substitution method) - the broader measure, not used as the AI denominator.
- IEA, Global Energy Review 2025 and Energy & AI 2025 (supply growth; data-center demand to 2030).

- Epoch AI, energy per query (~0.3 Wh median) and ML hardware energy efficiency (leading hardware ~40%/yr more efficient); IEEE Spectrum.
- OpenAI / TechCrunch (Feb 2026), ChatGPT ~900M weekly users.
- McKinsey, State of AI 2024/2025 (gen-AI adoption 65% to 72% of organizations).
- Gartner, >1B knowledge workers worldwide; enterprise agent adoption 25% to 50% by 2027.
- Uptime Institute, Global Data Center Survey 2024 (average PUE 1.56; hyperscale <=1.4).
- SpaceX / Falcon 9 (launch ~\$2,720/kg; Starship target <\$100/kg).
- Scientific American; Data Center Dynamics, space data centers (Starcloud, Google Project Suncatcher); launch-cost targets.

© 2026 Alessandro Golkar, Technical University of Munich. Interactive model and source: github.com/agolkar/ai-demand-model.
Feedback: golkar@tum.de. Generated by Claude Opus 4.8 (Anthropic) under supervision; undergoing verification.